

SHAPE AREA CALCULATOR

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1. INTRODUCTION

This project is a simple python program that can be used to calculate the area of various shapes. This program can be easily made using the python concepts taught in this course

2. PROBLEM STATEMENT

Calculating the area of shapes by hand is slow and easy to get wrong, especially for students and beginners who forget the formulas. This project is a menu-driven Python program that asks for the shape and its dimensions, then shows the area instantly. It also lets the user repeat calculations in one session without restarting.

3. OBJECTIVES

- Create a menu-driven Python applications
- Calculate the area of a square , rectangle , circle and python
- Use functions for modular programming
- Use conditional statements for menu selections
- Allow repeated calculations using a loop

4. FUNCTIONAL REQUIREMENTS

Square area: Take the side and calculate area using $\text{side} \times \text{side}$

Rectangle area: Take length and breadth and calculate area using $\text{length} \times \text{breadth}$

Triangle Area: Take perpendicular length and base and calculate the area using $0.5 \times \text{base} \times \text{height}$

Menu Area : Display choices , call the selected functions, show the result and ask whether to continue

5. NON-FUNCTIONAL REQUIREMENTS

- **Usability** : Simple prompts and menu for easy interaction
- **Maintainability**: Separate functions keep calculations organised
- **Reliability**: Correct results for valid numeric inputs.
- **Error Handling**: Invalid menu choices display an appropriate message

6. SYSTEM ARCHITECTURE

USER → SHAPE MENU → SHAPE SELECTION →
CORRESPONDING FUNCTION → INPUT DIMENSIONS → AREA
CALCULATION → DISPLAY RESULT → CONTINUE OR EXIT

7. PROCESS FLOW

START → DISPLAY WELCOME AND SHAPE MENU → SELECT
SHAPE(1-4) → CALL SELECTED FUNCTION → ENTER REQUIRED
DIMENSIONS → CALCULATE AREA → DISPLAY AREA →
CONTINUE (Y/N) → IF YES THEN CONTINUE OR ELSE STOP

8. DESIGN DECISIONS AND RATIONAL

The program defines functions for each supported shape . Each functions takes the required dimensions using input() , calculates the area and returns the result. The main program display a menu, accepts the user's choice , calls the appropriate functions , displays the area and asks whether the user wants to continue

9. TECHNOLOGIES USED

- PYTHON 3.X
- FUNCTIONS
- IF-ELIF-ELSE STATEMENTS
- WHILE LOOP
- BASIC ARITHMETIC OPERATIONS

10. SCREENSHOTS OF RESULTS

- MAIN MENU

```
===== RESTART: E:\COLLEGE\PYTHON\area calculation.py =====  
WELCOME  
please make a choice from the below listed shapes  
1.square  
2.rectangle  
3.circle  
4.triangle  
enter the number of your choice1
```

- SQUARE

```
enter the side of the square you want to calculate2  
the area of the shape is : 4  
do you want to continue?(y/n)
```

- RECTANGLE

```
enter the number of your choice2  
enter the length2  
enter the breadth3  
the area of the shape is : 6
```

- CIRCLE

```
enter the number of your choice3  
enter the radius7  
the area of the shape is : 153.86  
do you want to continue?(y/n)
```

- TRIANGLE

```
enter the number of your choice4  
enter the length of perpendicular2  
enter the length of base4  
the area of the shape is : 4.0  
do you want to continue?(y/n)
```

11. TESTING APPROACH

Test Case	Input	Expected Result
Square	Side = 2	Area = 4
Rectangle	Length = 2, Breadth = 3	Area = 6
Circle	Radius = 7	Area = 153.86
Triangle	Base = 4, Height = 2	Area = 4

12. CHALLENGES FACED

- Understanding function creation and function calls
- Passing user inputs into calculations
- Connecting menu choices with functions
- If-elif else statements.
- While loops
- Mathematical calculations
- Menu-driven programming

13. CHALLENGES FACED

- Understanding functions creation and function calls
- Passing user input into calculation
- Connecting menu choices with functions
- Using a while loop for repeated calculations
- Handling invalid menu choices

14. FUTURE ENHANCEMENTS

- Add more geometric shapes
- Add validation for negatives values.
- Use python's math module for more accurate value of pi
- Add perimeter calculation
- Add volume calculations for 3D shapes
- Create a graphical user interface
- Improve error handling

15. REFERENCES

VITyarthi – Build Your Own Project: General Project Instructions & Submission Guidelines. Python Essentials course concepts and classroom material.

16. CONCLUSION

The Shape Area Calculator demonstrates how basic Python concepts can be combined to create a useful console-based application. The project uses functions, user input, conditional statements, loops, and arithmetic operations to calculate the area of four geometric shapes. It provides a simple example of modular and menu-driven programming.